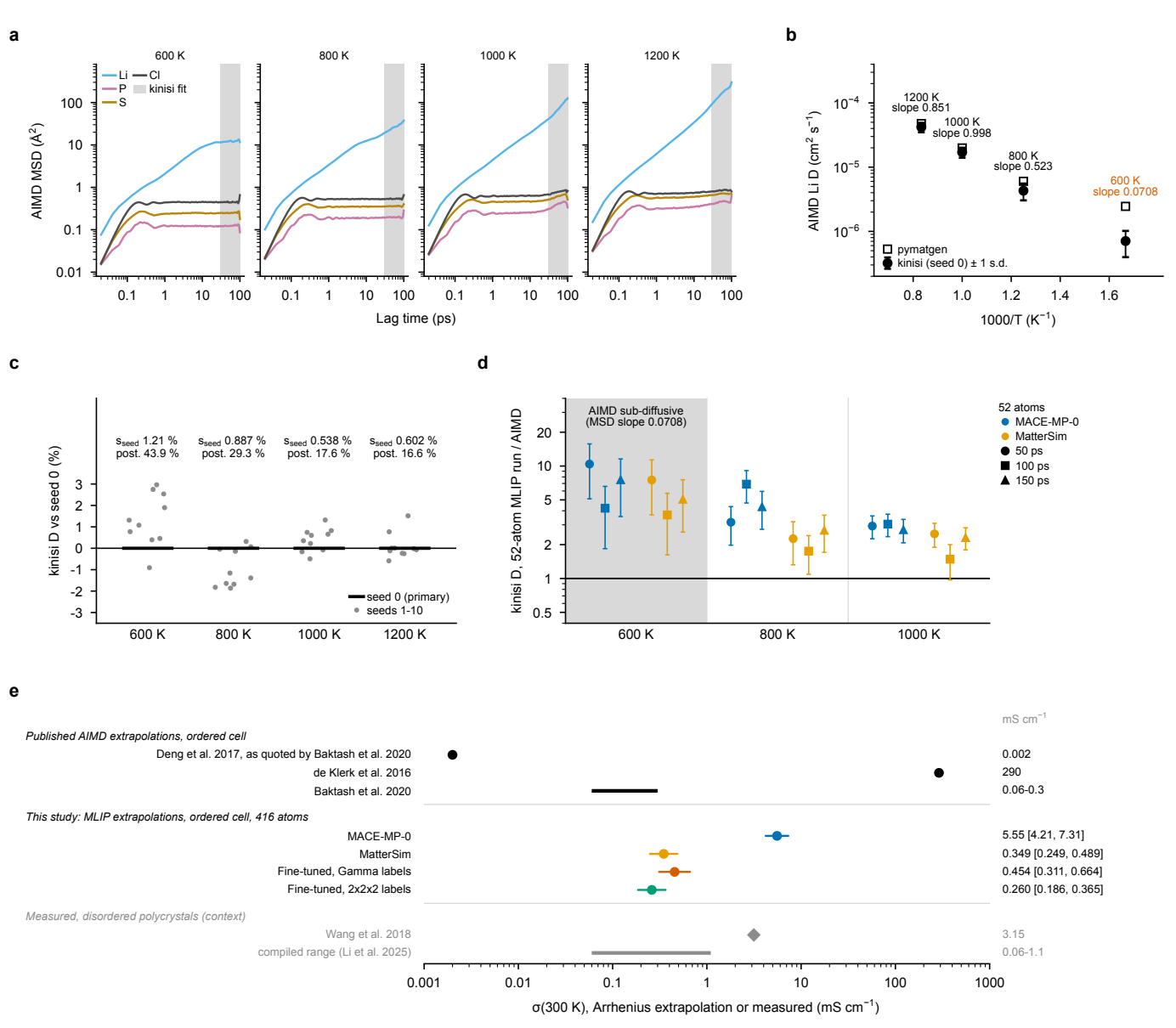

a
600 K
800 K
1000 K
1200 K
Li
Cl
P
kinisi fit
S
AIMD MSD (Å²)
Lag time (ps)
b
AIMD Li D (cm² s⁻¹)
1200 K
slope 0.851
1000 K
slope 0.998
800 K
slope 0.523
600 K
slope 0.0708
pymatgen
kinisi (seed 0) ± 1 s.d.
1000/T (K⁻¹)
c
kinisi D vs seed 0 (%)
s_seed 1.21 %
post. 43.9 %
s_seed 0.887 %
post. 29.3 %
s_seed 0.538 %
post. 17.6 %
s_seed 0.602 %
post. 16.6 %
seed 0 (primary)
seeds 1-10
d
kinisi D, 52-atom MLIP run / AIMD
AIMD sub-diffusive
(MSD slope 0.0708)
52 atoms
MACE-MP-0
MatterSim
50 ps
100 ps
150 ps
e
mS cm⁻¹
Published AIMD extrapolations, ordered cell
Deng et al. 2017, as quoted by Baktash et al. 2020
0.002
de Klerk et al. 2016
290
Baktash et al. 2020
0.06-0.3
This study: MLIP extrapolations, ordered cell, 416 atoms
MACE-MP-0
5.55 [4.21, 7.31]
MatterSim
0.349 [0.249, 0.489]
Fine-tuned, Gamma labels
0.454 [0.311, 0.664]
Fine-tuned, 2x2x2 labels
0.260 [0.186, 0.365]
Measured, disordered polycrystals (context)
Wang et al. 2018
3.15
compiled range (Li et al. 2025)
0.06-1.1
σ(300 K), Arrhenius extrapolation or measured (mS cm⁻¹)